Week 1 Objective: Identify literature and develop Outline/Structure

<table>
<tr><th>Activity</th><th>Status</th><th>Literature</th><th>Who?</th></tr>
<tr><td>Preliminary literature search on factors affecting fraud detection performance</td><td>Class imbalance
Concept drift and evolving fraud patterns
False positives
Real-time detection requirements
Explainability and regulatory compliance
Institutional and system heterogeneity
Data quality and feature availability
Feature engineering and feature selection
Computational complexity and scalability
Model overfitting and generalization
Privacy, security, and data governance constraints
Availability and representativeness of training data
Hyperparameter optimisation
Model interpretability
Adversarial attacks and fraud evasion techniques
Transaction volume and velocity
Changing customer behaviour
Integration with legacy banking systems
Cross-institutional data variability
Continuous model monitoring and retraining</td><td>Al-dahasi et al. (2025); Compagnino et al. (2025); Dichev, Zarkova and Angelov (2025); Faisal et al. (2024); Hashemi, Mirtaheri and Greco (2022); Hernandez Aros et al. (2024); Islam et al. (2024); Khalid et al. (2024); Khatib (2024); Li and Chen (2025); Luong and Xie (2026); Vanini et al. (2023); Zafar and Wu (2026)</td><td>Omatayo</td></tr>
<tr><td>Review literature on class imbalance</td><td>Class Imbalance
Nature and extent of class imbalance in fraud detection datasets
Impact of class imbalance on model performance
Effects on accuracy, precision, recall, and F1-score
Challenges in minority class prediction
Techniques Used to Improve Model Performance</td><td>Adebayo et al., 2026; Albalawi et al., 2025; Al-</td><td></td></tr>
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and techniques used to improve model performance	• Random oversampling • Random undersampling • Synthetic Minority Oversampling Technique (SMOTE) • Adaptive Synthetic Sampling (ADASYN) • Cost-sensitive learning • Class weighting • Ensemble learning methods • Threshold optimization • Feature engineering and feature selection • Hyperparameter tuning • Cross-validation techniques • Data augmentation methods	dahasi et al., 2025; Almazroi and Ayub, 2023; Boulieris et al., 2024; Cai and Yang, 2026; Chen, 2025; Compagnino et al., 2025; Luong and Xie, 2026.	Edem/Omotayo
Examination of the effects of concept drift and evolving fraud patterns on the effectiveness of machine learning fraud detection models. (And Adversarial Evolution)	• Concept Drift, Evolving Fraud Patterns, and Adversarial Evolution • Concept drift in fraud detection • Types of concept drift (sudden, gradual, incremental, recurring) • Evolution of fraud patterns and tactics • Behavioural changes in fraudsters • Adversarial evolution and fraud evasion techniques • Impact of concept drift on model accuracy and reliability • Model degradation over time • Continuous learning and model retraining • Incremental and online learning approaches • Adaptive and dynamic machine learning models • Ensemble updating techniques • Transfer learning for evolving fraud environments • Continuous model monitoring and performance evaluation • Best practices for maintaining long-term model effectiveness.	Adebayo et al., 2026; Cai and Yang, 2026; Compagnino et al., 2025; Gupta and Babu, 2025; Luong and Xie, 2026	Edem/Omotayo

Search and review literature on machine learning-based fraud detection systems	• Supervised machine learning fraud detection systems • Unsupervised machine learning fraud detection systems • Semi-supervised machine learning fraud detection systems • Reinforcement learning-based fraud detection systems • Deep learning-based fraud detection systems • Ensemble learning-based fraud detection systems • Hybrid machine learning fraud detection systems • Anomaly detection-based machine learning systems • Behavioural analytics-based machine learning systems • Real-time machine learning fraud detection systems • Graph-based machine learning fraud detection systems • Federated machine learning fraud detection systems • Explainable Artificial Intelligence (XAI)-based fraud detection systems • Adaptive and online learning fraud detection systems • Privacy-preserving machine learning fraud detection systems	AbouGrad and Sankuru (2025); Al-dahasi et al. (2025); Chen (2025); Compagnino et al. (2025); Dichev, Zarkova and Angelov (2025); Hashemi, Mirtaheri and Greco (2022); Islam et al. (2024); Khalid et al. (2024); Khatib (2024); Li and Chen (2025); Luong and Xie (2026); Zhang et al. (2023)	Sanni/Omatayo